EE603, Spring 2024
Final Project: Digital Color Restoration

Prepared by:
Marcus Chen
E-mail: mchen6446@sdsu.edu
Electrical and Computer Engineering
San Diego State University

Table of Contents

Table of Figures 2
Introduction 3
Objective function 5
Constraints 7
Results 7
Conclusion 12
References 15

Table of Figures

- Figure 1: RGB cartesian representation
- Figure 2: HSV cylindrical representation
- Figure 3: YCbCr plot of Cb and Cr color space
- Figure 4: Proof of concept reference and modified image
- Figure 5: Reconstruction of images using different spectrum bands
- Figure 6: Results on sampled images
- Figure 7: Results on less successful sampled images

Introduction

It is often said that a picture lasts forever, or that it is worth a thousand words, but unfortunately in reality that is not the case. In a world where multimedia streaming is the norm, it is often easy to overlook the importance that physical media still holds: all restorations and remasters to beloved films and television programs are done by running film through digital devices, often at improved resolution and pixel depth. While physical media often produces results that are better in quality to their digital counterparts (especially their digital counterparts from years prior), there are a variety of environmental and material factors that lead to the inevitability of decaying [1]. Media such as photographs and films are often much more suited to last for longer periods of time compared to something such as newsprint, but it is also not immune to the effects of light exposure and acid exposure.

Sailor Moon is an anime series developed by Toei Animation in the year 1992, lasting 200 episodes and becoming a quick cultural zeitgeist not only in Japan, but at an international scale [2]. Due to its warm and pastel color schemes and exquisite watercolor backgrounds, Sailor Moon is remembered fondly for its pink hues and its likeness to the city pop aesthetic, but the figurative rose-tinted lenses that it is remembered for might be due to a rose-colored decay [3]. As opposed to using the film negatives (the inverse of the original print colors), which are typically standard for image restoration, Toei opted to keep and utilize the film positives using Kodak Eastman color film, something well-known for having issues with color degradation [4]. Sailor Moon is not the only Toei property that has been affected by these issues: the original Dragon Ball film has degraded to a color green, providing a result that is much less pleasant to look at than the pink hues of Sailor Moon. Of course, there is still room for color ambiguity due to both the programs being designed for CRT television screens, which display light through exciting phosphorus instead of digital, which displays light as pixels [5].

For this project, optimization will be utilized in application to color restoration of a simulated degraded image using different spectrum bands of various frames. For simplicity, images will simply be restored to the colors of prior scans using the simplex method to find the optimal scalar modifiers for each of the bands. The image's restoration quality will be judged through both a qualitative and quantitative analysis: first by seeing if the restored image is visually like prior scans and then by looking at the value discovered of the objective function.

Decision Variables:

Many digital images utilize three spectrum bands in different color spaces: Red, Green, and Blue (RGB); Hue, Saturation, and Value (HSV); and Luminance, Blue-difference Chrominance, and Red-difference Chrominance (YCbCr).

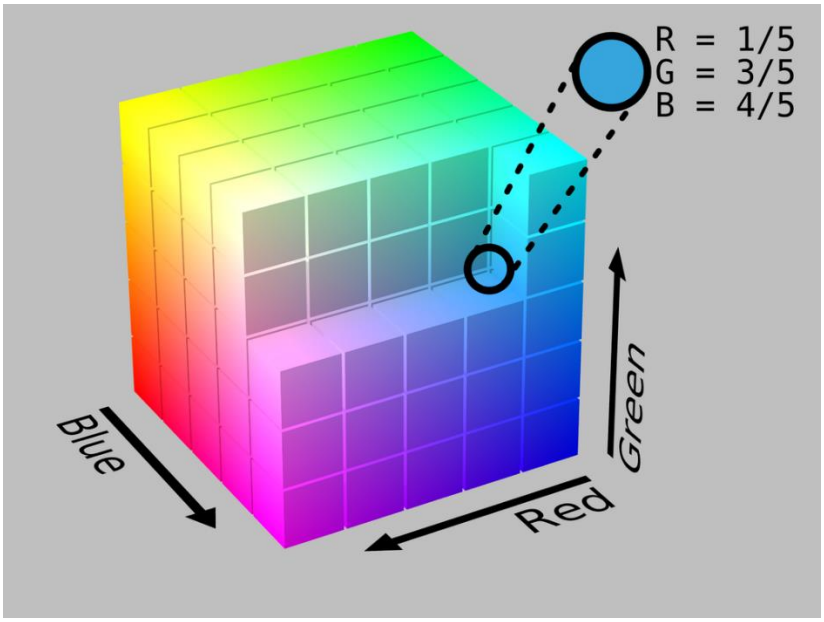


Figure 1: RGB Cartesian Representation [6]

The RGB color space translates all colors to permutations of three primary colors: red, green, and blue. While recent technology and displays have increased capabilities for pixel depth, the standard depth used is 8 bits, or binary digits of data, resulting in a range of each color from 0 to 255 or a total of 16,777,216 possible colors. At

values of 0, colors approach black and at values of 255, colors approach full intensity, with white being produced with all three color bands being at value 255. A counterpart to the RGB color space is the CMYK color space, which is the inverse and uses the primary colors cyan, magenta, and yellow (secondary colors in RGB); this is mostly used for printing since it is subtracted from the color white. Most imagers display using the RGB color space, so to see the results, the other color spaces will have to be subsequently translated to RGB.

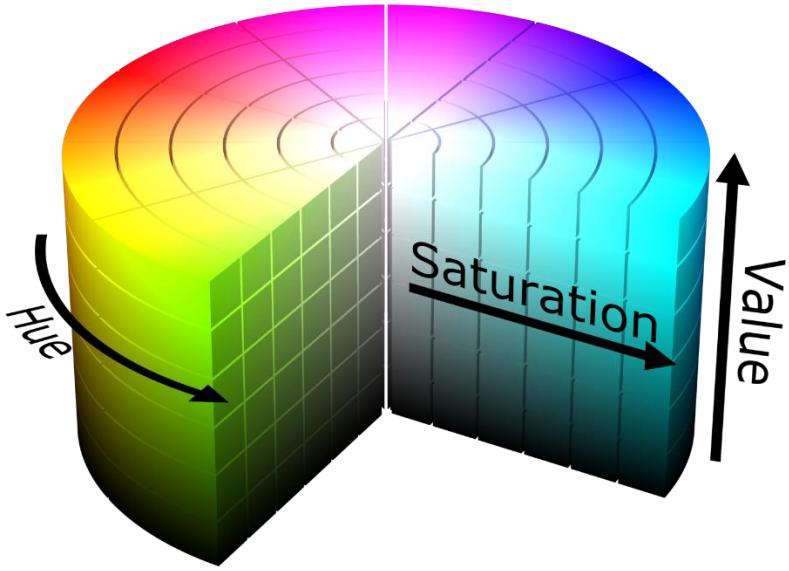


Figure 2: HSV Cylindrical Representation [7]

The HSV color space translates all colors to permutations of three characteristics: hue, saturation, and value. Instead of the RGB color space which uses a cartesian representation of color, HSV utilizes a cylindrical representation, with hue being based on the angular dimensions of the cylinder, saturation being based on the position along the radius of the cylinder, and value being based on the height of the cylinder. Hue refers to the position along a color wheel that a particular pixel exists on with red being valued at 0° or 360° point of the cylinder and green and blue being positioned on the 120° and 240° points of the cylinder respectively. Saturation refers to the intensity of a particular color, with a 0 being valued at gray and a 1 being valued at the maximum intensity of a particular color. Value refers to the brightness of a particular color, with 0 being black and 1 being the maximum possible luminous intensity.

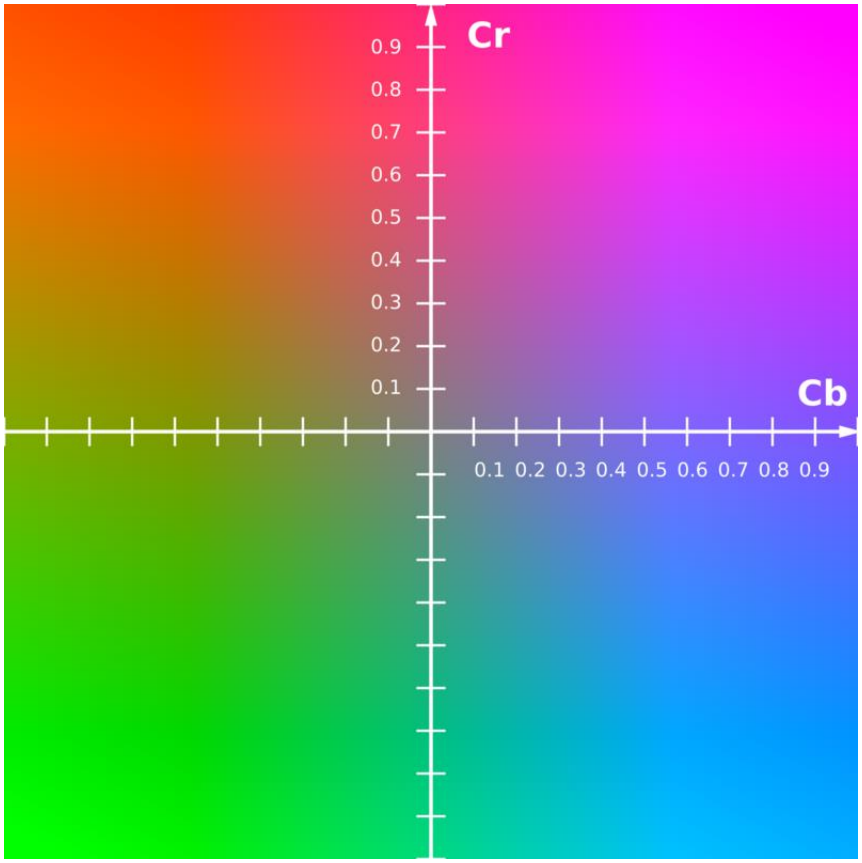


Figure 3: CbCr Plane of YCbCr color space [8]

The YCbCr color space translates all colors to permutations of the luminance, or brightness, component and the two chrominance, or color, components: red-difference and blue-difference. Similar to RGB, the intensity is typically perceived to be around a pixel depth of 8 bits, but it is often useful for image and video compression due to being able to reduce resolution while maintaining the likeness of a particular image. The luminance component is the intensity of the entire image according to brightness and often can be displayed as a grayscale version of a color image. Chrominance red-difference looks at the intensity of red in an image and chrominance blue-difference looks at the intensity of blue in an image; it is often down sampled for usage of image compression.

For the project, the decision variables will be based off of scalar modifications to the aforementioned parameters in the different color spaces: d_R, d_G, and d_G for RGB; d_H, d_S, and d_V for HSV; and d_Y, d_{Cb}, and d_{Cr} for YCbCr. The resulting modified image will appear as:

- For the RGB model: $I_{RGB}{}' = [d_R I_R, d_G I_G, d_B I_B]^T$
- For the HSV model: $I_{HSV}{}' = [d_H I_H, d_S I_S, d_V I_V]^T$
- For the YCbCr model: $I_{YCbCr}{}' = [d_Y I_Y, d_{Cb} I_{Cb}, d_{Cr} I_{Cr}]^T$

Objective function

The objective function will be based on a minimization of the difference between a prior image interpolated and a current image with the scalar modifications di at each color band. In order to convert the optimization problem to a linear format, the equation used is the Delta E (CIE 2000) formula, to compare colors from a prior and reconstructed image [9]. CIE 2000 is defined by:

$$E_{00}^{*} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \frac{\Delta C'}{k_C S_C} \frac{\Delta H'}{k_H S_H}}$$

where:

$$\Delta L' = {L_2}^* - {L_1}^*, \qquad \bar{L} = \frac{{L_1}^* + {L_2}^*}{2}, \qquad \bar{C} = \frac{{C_1}^* + {C_2}^*}{2}, \qquad {C_i}^* = \sqrt{{a_i}^{*2} + {b_i}^{*2}}, i = 1,2$$

$$a_i{}' = {a_i}^* + \frac{{a_i}^*}{2}\left(1 - \sqrt{\frac{\bar{C}^7}{\bar{C}^7 + 25^7}}\right), i = 1,2$$

$$\bar{C}' = \frac{{C'_1}^2 + {C'_2}^2}{2}, \qquad \Delta C' = C'_2 - C'_1, \qquad C'_i = \sqrt{{a'_i}^2 + {b_i}^{*2}}, i = 1,2$$

$$h'_i = atan2({b_i}^*, a'_i)\ mod\ 360°, i = 1,2$$

$$\Delta h' = \begin{cases} h'_2 - h'_1, |h'_2 - h'_1| \leq 180° \\ h'_2 - h'_1 + 360°, |h'_2 - h'_1| > 180°, h'_1 + h'_2 < 360° \\ h'_2 - h'_1 - 360°, |h'_2 - h'_1| > 180°, h'_1 + h'_2 \geq 360° \end{cases}$$

$$\Delta H' = 2\sqrt{C'_1 C'_2} sin(\Delta h'/2), \qquad \bar{H}' = \frac{\Delta h'}{2}$$

$$T = 1 - 0.17cos(\bar{H}' - 30°) + 0.24cos(2\bar{H}') + 0.32cos(3\bar{H}' - 6°) - 0.20cos(4\bar{H}' - 63°)$$

$$S_L = 1 + \frac{0.015(\bar{L} - 50)^2}{\sqrt{20 + (\bar{L} - 50)^2}}, \qquad S_C = 1 + 0045\bar{C}', \qquad S_H = 1 + 0.015\bar{C}'T$$

$$R_T = -2\sqrt{\frac{\bar{C}^7}{\bar{C}^7 + 25^7}} sin\left[60° \cdot exp\left(-\left[\frac{H' - 275°}{25°}\right]^2\right)\right]$$

and k_L, k_C, and k_H are usually unity. The conversion from RGB to the L*a*b format is defined by:

$$\begin{aligned} L^* &= 116f(Y/Y_n) - 16 \\ a^* &= 500(f(X/X_n) - f(Y/Y_n)) \\ b^* &= 200(f(Y/Y_n) - f(Z/Z_n)) \end{aligned}$$

where

$$f(t) = \begin{cases} \sqrt[3]{t}, t > \delta^3 \\ \frac{1}{3}\mathrm{t}\delta^{-2} + \frac{4}{29}, \mathrm{t} \leq \delta^3 \\ \delta = \frac{6}{29} \end{cases}$$

and

$$C_{lin} = \begin{cases} \frac{C_{RGB}/255}{12.92}, (C_{RGB}/255) \leq 0.04045 \\ \left(\frac{(C_{RGB}/255) + 0.055}{1.055}\right)^{2.4}, (C_{RGB}/255) > 0.04045 \end{cases}$$

for each color R, G, and B

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.4124 & 0.3576 & 0.1805 \\ 0.2126 & 0.7152 & 0.0722 \\ 0.0193 & 0.1192 & 0.9505 \end{bmatrix} * \begin{bmatrix} R_{lin} \\ G_{lin} \\ B_{lin} \end{bmatrix}$$

for the pixels in each image [10]. The formulas subscripts for the objective function analyze the difference between the prior and reconstructed image and will be scaled according to a

$$\begin{bmatrix} d_0 & d_1 & d_2 \\ 1 & 1 & 1 \end{bmatrix}$$

model with the top row representing the spectrum scaling to reconstructed image and the scaling to the prior reference image.

The objective function will therefore be described as

$$min\{\sum_{m=0}^{m=M-1}\sum_{n=0}^{n=N-1} E_{00}^{*}(I_{m.n}{}', I_{m,n})\}$$

where I' represents the image with the modified colors and I represents the prior image upscaled through bilinear interpolation to be the same size as I' both of which contain the color space data for each pixel.

Thankfully the aforementioned math can be solved using Python's OpenCV and color science libraries using the function `cv2.cvtColor(image, cv2.COLOR_RGB2Lab)` and `colour.delta_E(image1_lab, image2_lab, method='CIE 2000')` respectively [11]. Python code is attached to the project submission.

To perform the minimization, SciPy's minimization function is used using `scipy.optimize.minimize(func, x0, method='Nelder-Mead', constraints={constraints})` [12]. The Nelder-Mead optimization method was chosen because it is an adaptation of the simplex method for the optimization of a somewhat more complex space.

Constraints

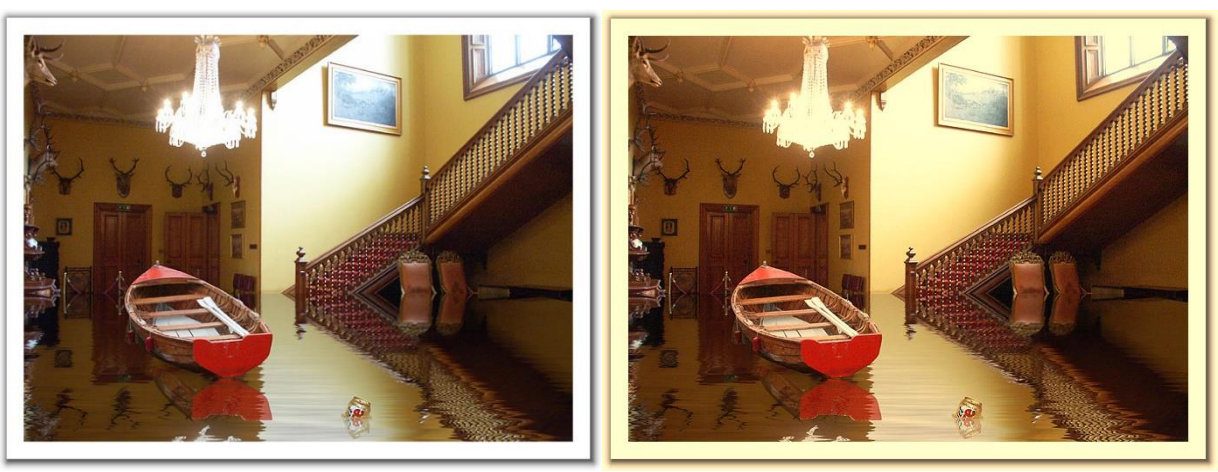
Because three models are being developed for the objective function, the main difference originates from the constraints given to images on each color space before they are converted to L*a*b* format for the objective function. Since pixels have a set luminance value strictly between the values of 0 and 255, there is a wrap-around effect if a value goes beyond the 8 bit limitation; to prevent that, in lieu of using standard constraints, the decision variables modified into the objective function will instead be sent through a truncation between the values of 0 and 255 for the R, G, B, S, V, Y, Cb, and Cr color spaces and between 0 and 179 for the H color space.

As a result, the constraints to each pixel will appear as:

$$P_{x,y}{}' = \begin{cases} 0, P_{x,y}d_i < 0 \\ 255, P_{x,y}d_i > 255 \\ P_{x,y}d_i, otherwise \end{cases}$$

and will be truncated before being sent into the objective function.

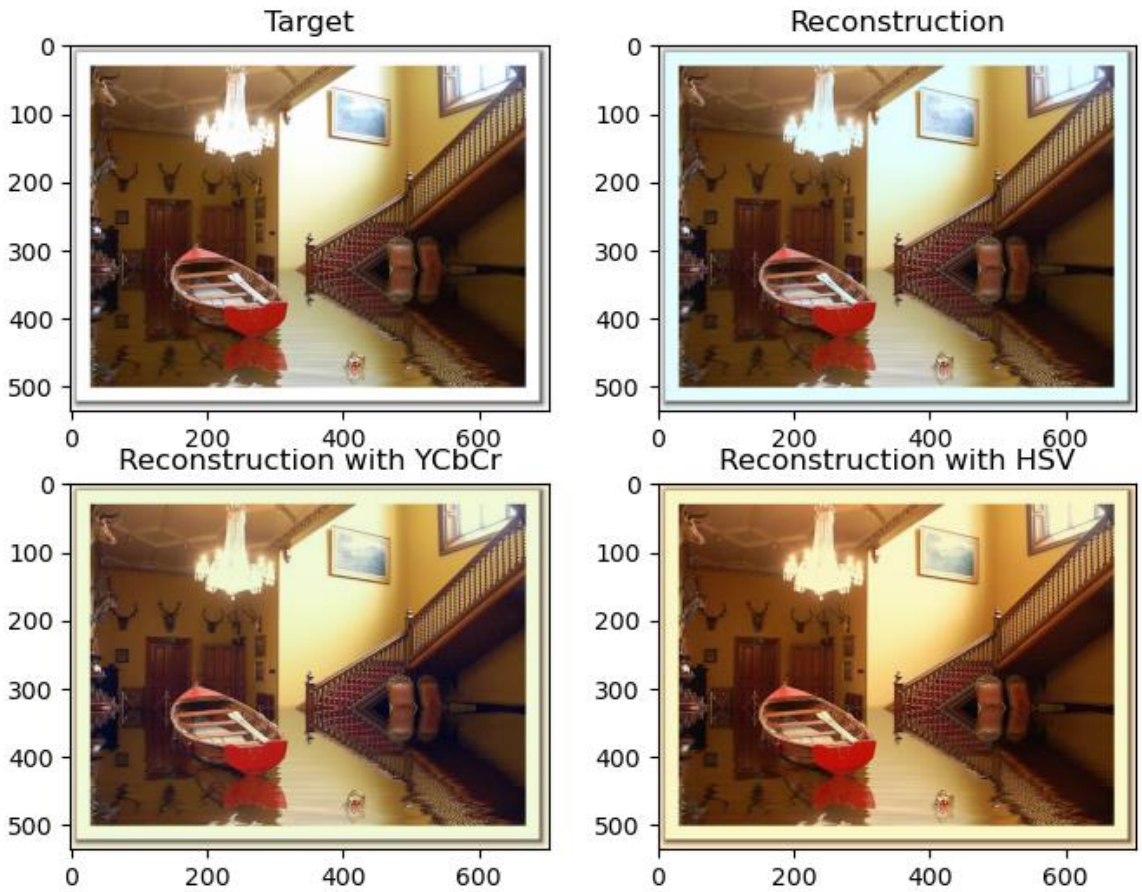
Results



a, b

Figure 4: (a) Proof of concept reference image, (b) Proof of concept modified image

To test out the proof of concept for the linear model, I used the sample image as well as one I modified in color using an image editor over my phone; by default, the summation of the delta E function for every pixel was valued at 1720090.3272: the ground zero of the experimentation (Fig. 4). The goal will be to return the image to something that resembles an item as close to the unmodified image as possible using the three different kinds of color spectrum bands.

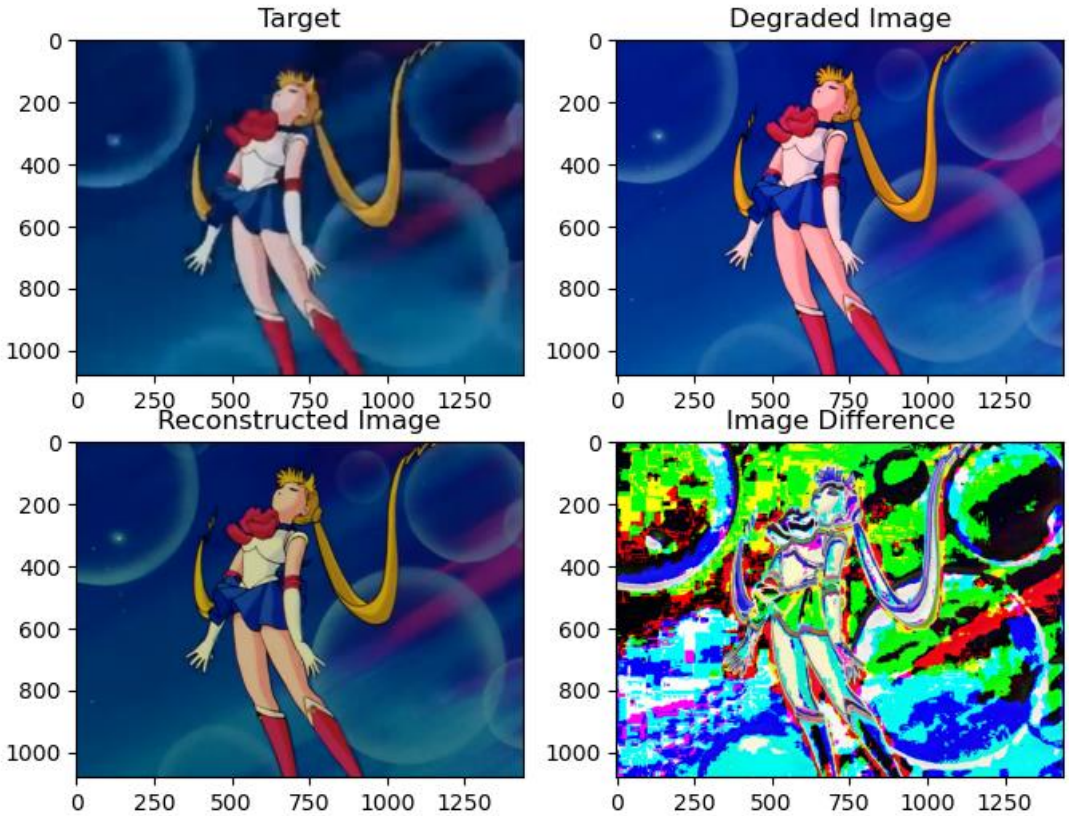


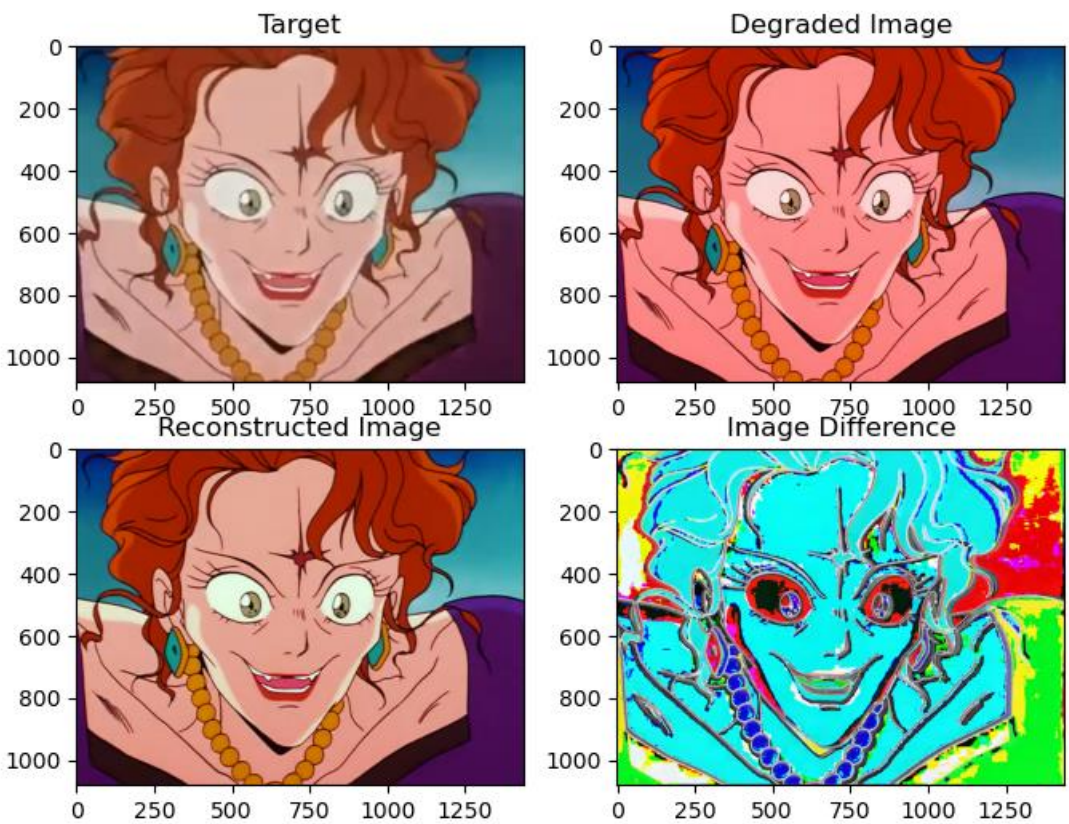
a, b
c, d

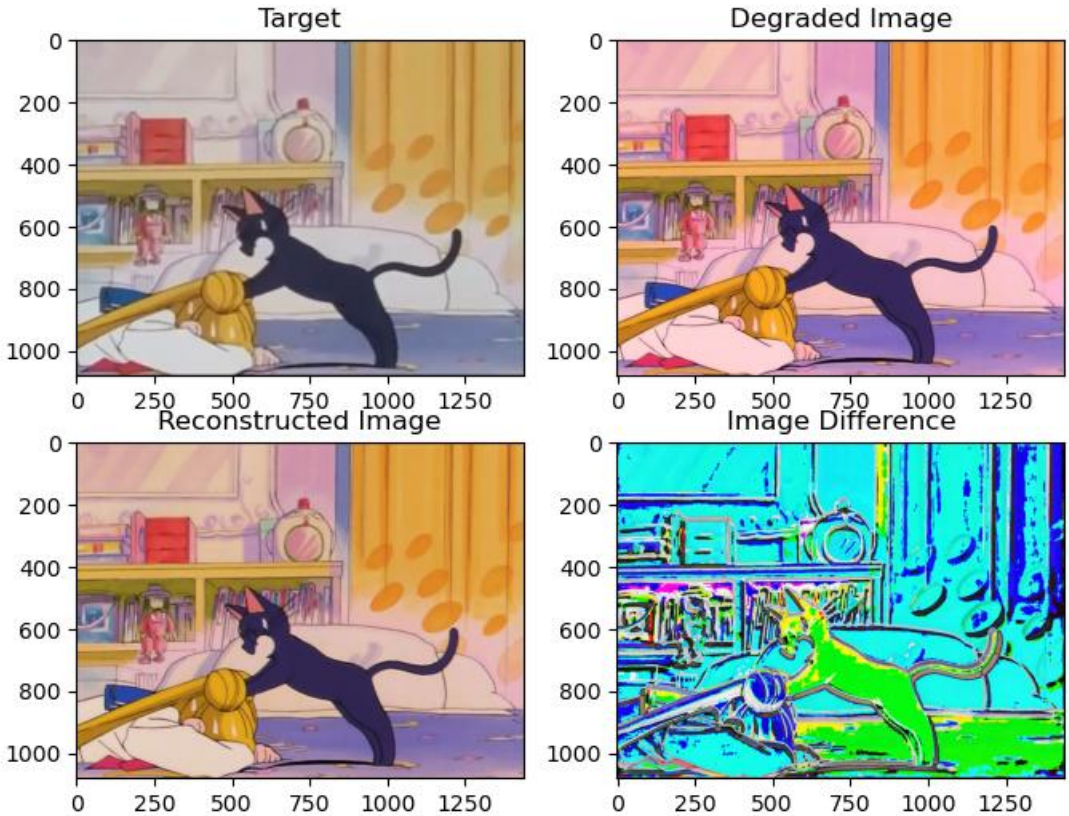
Figure 5: (a) Reference image, (b) Image reconstructed from the RGB color spectrum, (c) Image reconstructed from the YCbCr color spectrum, (d) Image reconstructed from the HSV color spectrum

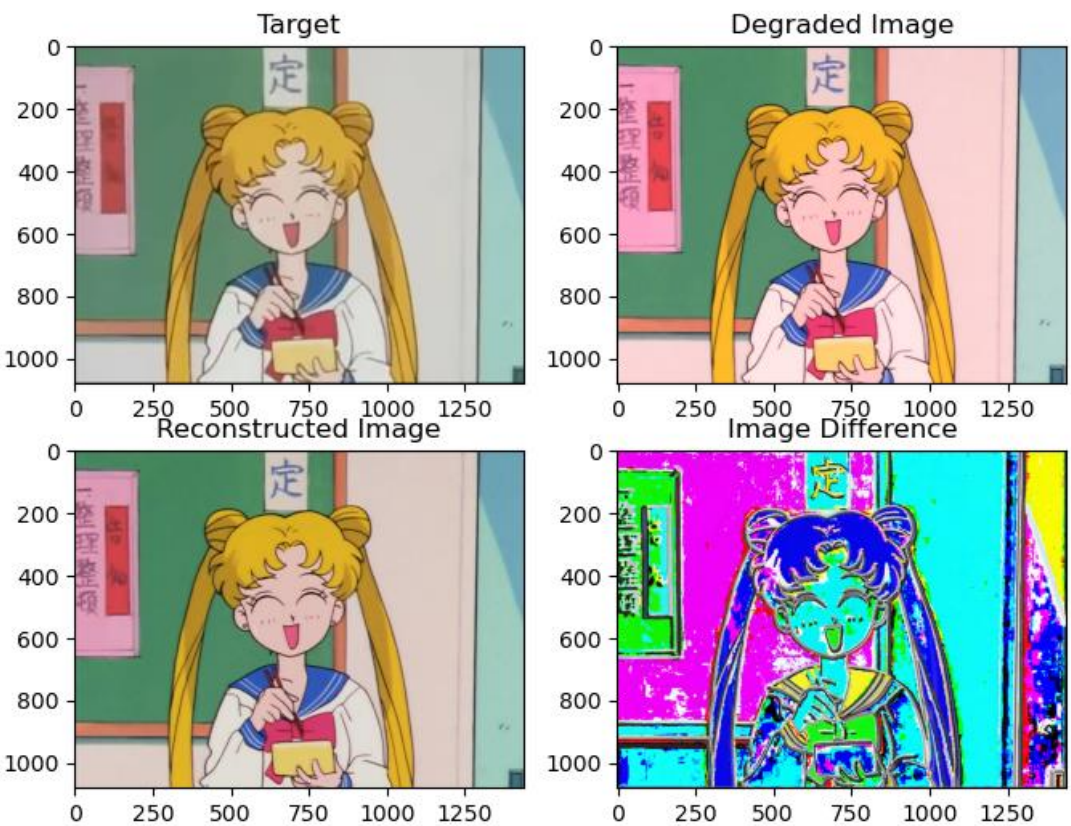
While the other color spaces hold their own individual advantages, it is ultimately RGB that resulted in the closest image due to the objective function in L*a*b* format most easily translating to the RGB color space. Additionally, compared to the original image CIE2000 difference, the one that is farthest from the value is the RGB value as well. The summation of the delta E value for the RGB bands was 688841.074936, or a 60% improvement, while the summation of the delta E values for the YCbCr and HSV bands were 1269935.20882 and 1405208.79925, only a 26% and 18% improvement respectively.

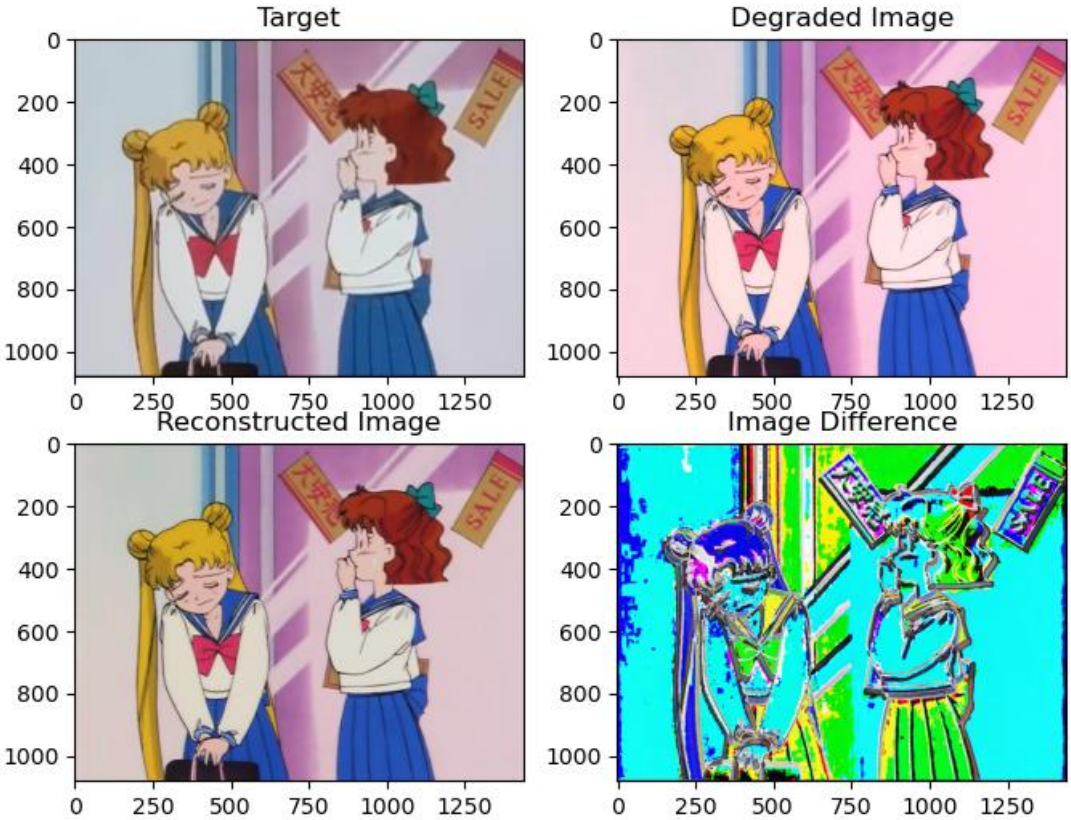
Based on the results of the different color space, RGB was used for the rest of the experiments over screencaps of Pretty Guardian: Sailor Moon episode 1 [2].


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250


Target
Degraded Image
Reconstructed Image
Image Difference
定
0
200
400
600
800
1000
0
250
500
750
1000
1250


Target
Degraded Image
Reconstructed Image
Image Difference
SALE
0
200
400
600
800
1000
0
250
500
750
1000
1250

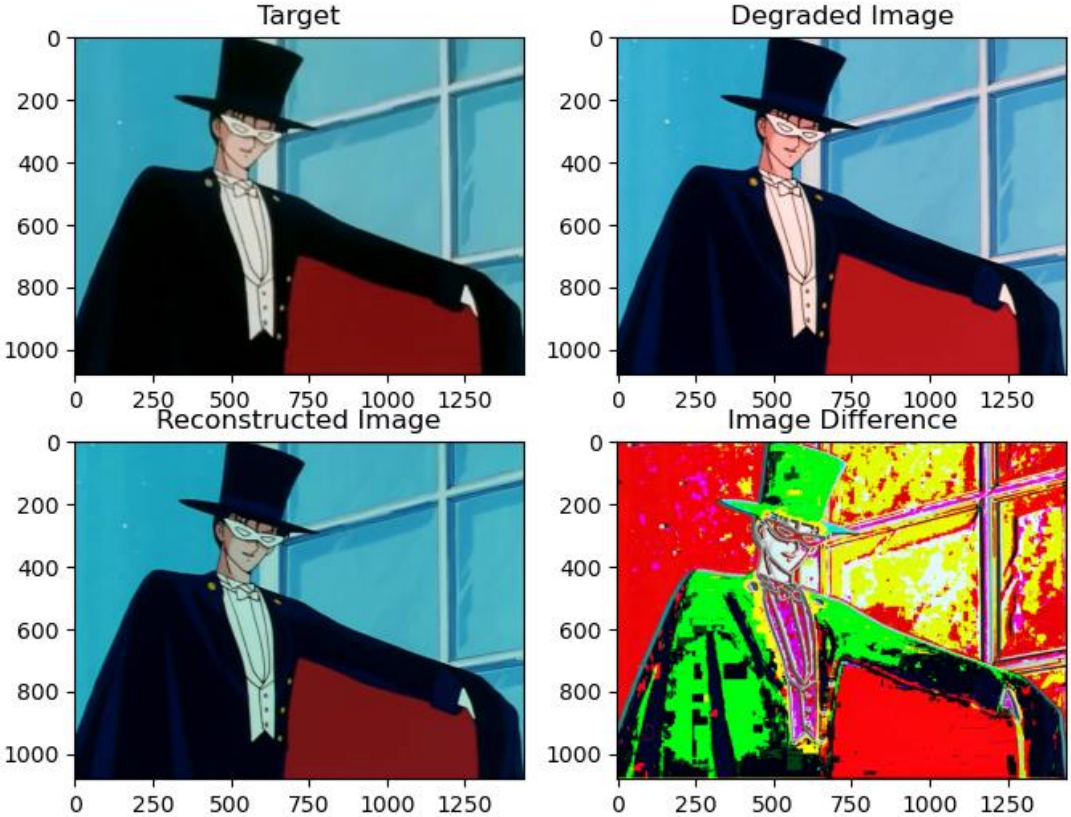

Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250

Figure 6: Results of RGB band reconstruction using optimization model [2]

While the initial plan was to find an average set of decision variables to use over all the screencaps, images unfortunately do not linearly decay either. The resulting decision variable weights, original delta E results, new delta E results, and percentage of improvement are depicted as such:

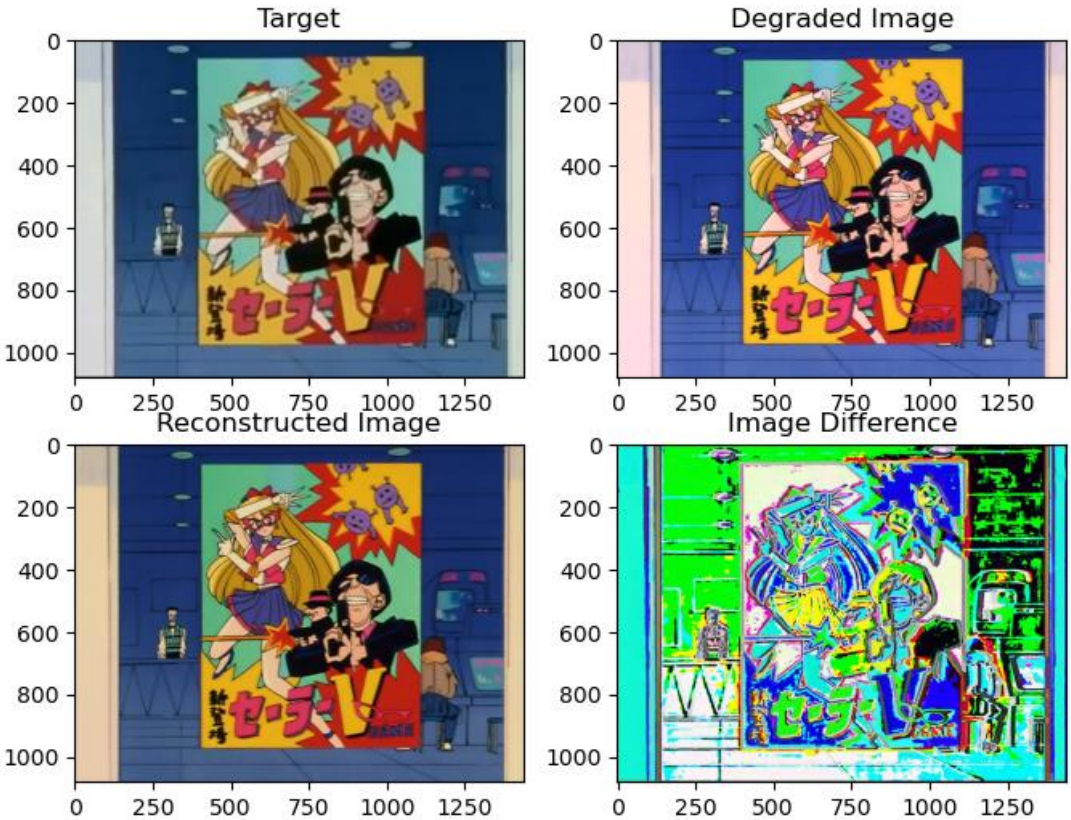
Frame	Weights	Original Delta E	New Delta E	% Improvement
1	[0.79217436 0.98040873 0.72790101]	24536468.38	14353988.05	41.50%
2	[0.91256059 1.16861554 1.05917191]	19089210.99	16502820.68	13.55%
3	[0.90921162 0.98743723 0.90100397]	15508966.79	13265572.21	14.47%
4	[0.88787144 0.96214723 0.91436317]	13785908.12	11078952.46	19.64%
5	[0.86945657 0.95714235 0.89454372]	16893338.66	12857371.25	23.89%
6*	[0.91209662 0.91754758 0.76051037]	22802316.48	18045120.06	20.86%
7*	[1.14883503 0.88248345 0.74120477]	14962440.53	10819630.11	27.69%
8	[0.65921003 0.97459486 0.93325614]	14579397.28	9134499.865	37.35%

All resulting Delta E 2000 computations for original and reconstructed images

Even though the percentage of improvement seems to be less significant on some compared to others, the resulting changes of weight produced images that have an overall lower color difference compared to their original counterparts, especially qualitatively. One notable improvement is the color of the backgrounds notably in image 1 and 8, where the dominant blues are much more closer to the initial coloration.

Conclusion

While this project was mostly successful in being able to restore some of the original coloring of various screencaps, it is not without its own issues.


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250

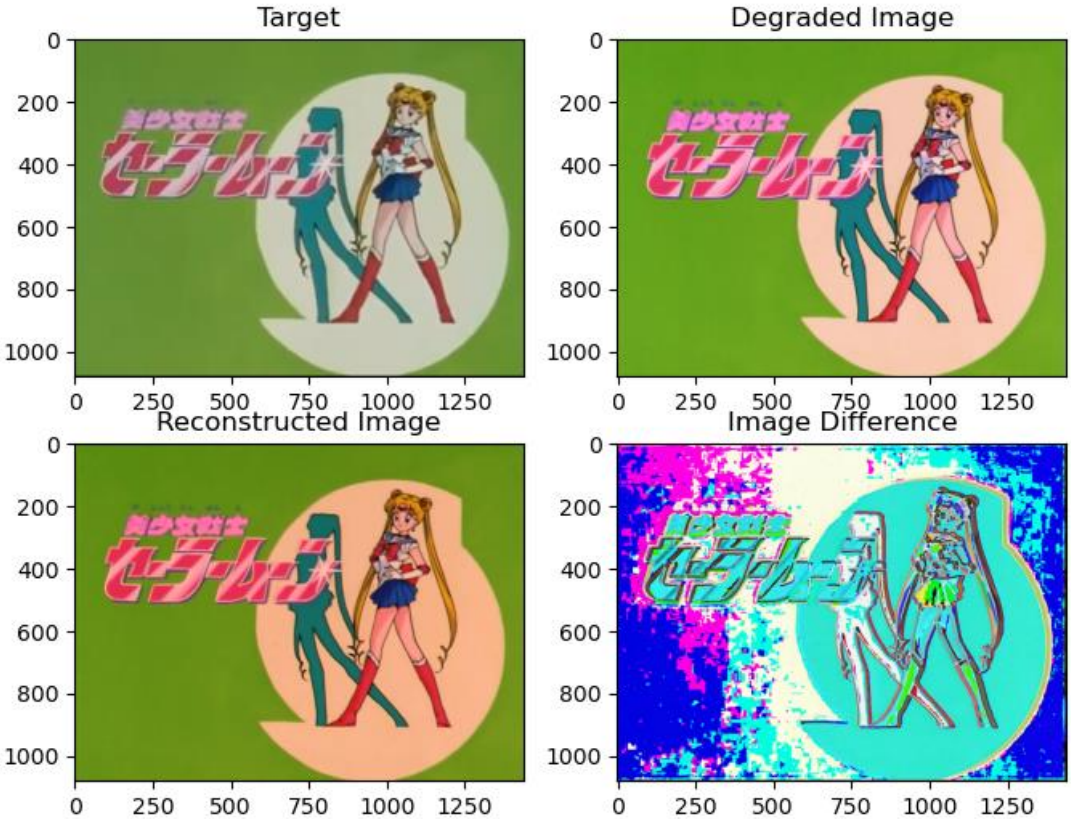

Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250

Figure 7: Less successful results of RGB image band reconstruction using optimization model [2]

What is often algorithmically the most optimal method to restore colors does not always match what is typically perceived psychovisually as the most important feature of an image. The latter of these two images has a reconstructed delta E that is lower than almost all the images above save the last one and a higher percentage of improvement, but it is qualitatively in a worse state than before it was re-weighted. Within a typical human perception of the image, the most prominent features are the bright images amidst a colder tone background, but the algorithm prioritizes the background colors as they are the dominant aspect spatially. Despite that, there are methods to provide these priorities through the process of edge detection, or a high pass filter that finds the edges of an image.

Through this project, a linear programming optimization model proved useful for performing color correction to a series of images that have been modified or degraded due to age. While digital media often has a longer shelf life than their physical counterparts, it is also not immune to gradual degradation; as images transfer across devices and encounter new image display standards, there is often a compression scheme that occurs to reduce the amount of data required to view an image. Lossless compression exists to display legal documents and medical diagrams, but it is often that standards like lossy JPEG are often the standard for image displays [13]. The process of quantization that occurs to reduce psychovisual redundancy results in some data becoming irrecoverable when it is decoded; while it is often not immediate at first, it becomes a larger issue when transferring to different hardware platforms. For both physical and digital media, it is important to adopt safe preservation habits such as keeping media in places safe from environmental impacts, using film negatives, and transferring data when its media is facing signs of aging; for media that are facing issues, optimization can be used to find out what its original appearance can be.

References

[1] E. Smith, "Luster Lost," Tedium, 2 March 2024. [Online]. Available: https://tedium.co/2024/03/02/why-physical-objects-decay/.

[2] J. Sato, Director, *Pretty Soldier Sailor Moon.* [Film]. Japan: Toei Animation, 1992.

[3] K. Mather, "'Sailor Moon' fans are surprised to learn the reason behind the anime's aesthetic pink hue: 'It wasn't supposed to look like that'," Yahoo! News, 23 January 2024. [Online]. Available: https://news.yahoo.com/sailor-moon-fans-are-surprised-to-learn-the-reason-behind-the-animes-aesthetic-pink-hue-it-wasnt-supposed-to-look-like-that-005916237.html.

[4] A. Amidon, "Film Preservation 101: Why are old films sometimes pink?," National Archives Special Media Division, 11 January 20116. [Online]. Available: https://unwritten-record.blogs.archives.gov/2016/01/11/film-preservation-101-why-are-old-films-sometimes-pink/.

[5] "Canon Science Lab," Canon Global, [Online]. Available: https://global.canon/en/technology/s_labo/light/002/03.html.

[6] SharkD, Artist, *RGB-Cube.* [Art]. Wikimedia Commons, 2010.

[7] SharkD, Artist, *HSV color solid cylinder.* [Art]. Wikimedia Commons, 2015.

[8] S. A. Eugster, Artist, *YCbCr-CbCr Scaled Y50.* [Art]. Wikimedia Commons, 2010.

[9] G. Sharma, W. Wu and E. N. Dalal, "The CIEDE2000 Color-Difference Formula: Implementation Notes, Supplementary Test Data, and Mathematical Observations," *Color Research and Application,* vol. 30, no. 1, pp. 21-30, 2005.

[10] "Delta E (CIE 2000)," Bruce Justin Lindbloom, 8 April 2017. [Online]. Available: http://www.brucelindbloom.com/index.html?Eqn_DeltaE_CIE2000.html.

[11] "colour.delta_E," Colour Developers, 2013. [Online]. Available: https://colour.readthedocs.io/en/latest/generated/colour.delta_E.html.

[12] "scipy.optimize.minimize," The SciPy community, 2008. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.minimize.html.

[13] D. R. Bull and F. Zhang, "Chapter 4 - Digital picture formats and representations," in *Intelligent Video and Image Compression (Second Edition)*, Academic Press, 2021, pp. 107-142.